

Here is the zoomed-in, highly detailed Gantt chart for **Phase 1: The Data & ML Foundation (Weeks 1 to 10)**.

To get that 10 LPA job, you must treat this phase like building the foundation of a skyscraper. If your Pandas and Math skills are weak, you will completely fail when trying to build advanced AI Agents later.

Detailed Phase 1 Gantt Chart (Weeks 1 - 10)

Legend:  = Active Learning/Building |  = Practice/Review

Week	Focus Area	Detailed Topics	Core Process to Understand (The "Why")	W1	W2	W3	W4	W5	W6	W7	W8	W9
W1	Python & Git	OOP, Data Structures (Lists/Dicts), Virtual Envs, Git (Commit/Push/Branch)	Process: How to write modular code. PyTorch uses Object-Oriented Programming heavily. Git is mandatory for tech teams.									
W2	SQL & Linux	Basic CLI (<code>cd</code> , <code>ls</code> , <code>grep</code>), SQL Joins, Group By, Window Functions, CTEs	Process: How to extract data from company databases. AI models need data; SQL is how you get it.									
W3	Linear Algebra	Vectors, Matrices, Dot Products, Matrix Multiplication	Process: AI models are just giant matrices multiplying each other. You need to know how data shapes change.									
W4	Prob & Stats	Mean/Variance, Normal Distribution, Probability, Bayes Theorem	Process: How to deal with uncertainty. ML is just statistics disguised as code.									
W5	NumPy & Pandas	Arrays, Broadcasting, DataFrames, Missing Value handling, Filtering	Process: How to clean dirty data. 80% of an ML Engineer's job is fixing bad data before training.									

W6	EDA & Viz	Matplotlib, Seaborn, Feature Engineering, Outlier Detection	Process: How to "look" at data to find patterns. If garbage goes into an AI, garbage comes out.	////	////	////	////	////	■	////	////	////
W7	Supervised ML	Linear/Logistic Regression, Random Forests, XGBoost	Process: How to predict numbers (Regression) or categories (Classification) using labeled data.	////	////	////	////	////	////	■	////	////
W8	Unsupervised & Eval	K-Means Clustering, PCA. Metrics: F1-Score, ROC-AUC, Precision/Recall	Process: How to evaluate if your AI is actually smart or just guessing. Cross-validation to prevent cheating.	////	////	////	////	////	////	////	■	////
W9	Project Part 1	Data Scraping, Cleaning, Feature Engineering, Model Training	Process: Building the backend pipeline of a real-world product.	////	////	////	////	////	////	////	////	■
W10	Project Part 2	UI (Streamlit), Deployment (HuggingFace Spaces), GitHub README	Process: Showing your work to recruiters. A model on your laptop is useless; it must be on the internet.	////	////	////	////	////	////	////	////	////

Let's Start Step 1 NOW: Python & Git (Week 1)

Forget theory for a minute. You are going to set up your environment like a professional engineer. No more simple "Jupyter Notebooks" running blindly.

Your Goal for the Next 7 Days:

Become comfortable with Python's Object-Oriented Programming (OOP) and version control (Git).

Day 1 & 2: The Pro Setup & Git Basics

- Action:**

1. Download and install **VS Code** (not PyCharm, VS Code is industry standard for AI right now).
2. Install **Git** on your machine.

3. Create a GitHub account.

- **Learn:** Open your terminal (Command Prompt/Mac Terminal) and learn: `cd` , `mkdir` , `ls` (or `dir`).
- **Git Process to understand:** `git clone` (download code), `git add .` (stage changes), `git commit -m "message"` (save changes locally), `git push` (send to GitHub).
- **Task:** Create a repo on GitHub called `AI-10LPA-Journey` , clone it to your PC, add a text file saying "Day 1 complete", and push it back to GitHub.

Day 3 & 4: Python Virtual Environments & Packages

- **The Process to Understand:** *Dependency Hell*. If Project A needs Python 3.8 and Project B needs Python 3.10, your computer will break. Virtual environments isolate projects.
- **Action:**
 1. Open terminal in your VS Code.
 2. Type: `python -m venv myenv` (This creates a virtual environment).
 3. Activate it: `myenv\Scripts\activate` (Windows) or `source myenv/bin/activate` (Mac/Linux).
 4. Install a package: `pip install requests` .

Day 5 & 6: Python Object-Oriented Programming (OOP)

- **The Process to Understand:** AI libraries like PyTorch are built entirely on "Classes" and "Objects". You must know what `__init__` , `self` , and inheritance mean.
- **Action:** Watch a tutorial on Python OOP (Corey Schafer's OOP playlist on YouTube is the best).
- **Task:** Write a Python script with a `Class` called `DataCleaner` . Give it methods (functions) to lowercase a string and remove punctuation.

Day 7: The Mini-Integration Challenge

Combine everything you learned this week.

1. Activate your virtual environment.
2. Write a Python script using OOP that fetches a joke from a free public API (using the `requests` library).
3. Push this script to your GitHub repository using Git commands in the terminal.

Are you ready to do Days 1 & 2 today? Let me know if you need help installing VS Code/Git, or if you want the exact terminal commands to run!